

Smart Traffic Lights Using Local AI for Emergency Vehicle Priority

H gnanaprasanna

Department of Computer Science and Engineering

Data Science

Ballari Institute of Technology and Management

Ballari, Karnataka, India

Email:hgnanaprasanna@gmail.com

Abstract—Traffic congestion at signalized intersections can increase travel time and create difficulties for emergency vehicles such as ambulances and fire engines. Conventional fixed-time traffic signals are based mainly on predetermined schedules and cannot continuously respond to changing traffic conditions. This paper presents a conceptual smart traffic-light architecture that uses camera-based traffic information and local computing to support adaptive signal decisions. Local processing can reduce the amount of data that must be transmitted to a remote server and can support rapid decisions at the road intersection. The proposed architecture combines street cameras, smart pole computers, optional cloud storage, and traffic-light controllers. A dedicated emergency-vehicle priority function is included to coordinate nearby signals and provide a clear route. The literature supports adaptive traffic-signal control using machine vision and intelligent methods, while previous research has also examined emergency-vehicle-centered signal control. The proposed system is intended as a conceptual design for further implementation and evaluation.

Index Terms—smart traffic lights, local AI, machine vision, adaptive traffic signal control, emergency vehicles, intelligent transportation systems

I. INTRODUCTION

Traffic congestion is a major challenge for urban transportation systems. Conventional traffic signals commonly operate according to predetermined timing plans. Such approaches can work for predictable traffic patterns, but traffic demand can change rapidly because of queues, incidents, weather, or emergency situations. Intelligent traffic-signal control attempts to use information about current traffic conditions to improve signal operation.

Research on deep reinforcement learning has investigated intelligent traffic-signal control and identified traffic-signal control as a suitable application for learning-based methods [1]. Machine-vision research has also demonstrated the use of camera-based vehicle information for adaptive traffic-light control [2]. These studies indicate that traffic information obtained from road-side cameras can be used as an input to adaptive traffic-signal systems.

Emergency vehicles create a particularly important requirement. At congested intersections, an ambulance or other emergency vehicle may be delayed even when it is permitted to pass through a red signal. Previous research has proposed emergency-vehicle-centered traffic signal control to reduce

emergency-vehicle delay while limiting disruption to traffic in conflicting directions [3]. Based on these research directions, this paper proposes a conceptual local-AI traffic-light architecture with an emergency-vehicle priority function.

II. LITERATURE REVIEW

Rasheed et al. presented a review of deep reinforcement learning approaches for traffic signal control. Their work discusses traffic-signal-control attributes, deep reinforcement learning architectures, simulation platforms, complexity, design considerations, and open research issues [1].

Gorodokin et al. investigated adaptive traffic-light control using machine vision. Their work used traffic-monitoring cameras and vehicle detection to obtain traffic-flow information and support dynamic adjustment of signal phases and cycle durations [2]. This work supports the use of camera-based traffic information in an adaptive traffic-signal architecture.

Cao et al. proposed an emergency-vehicle-centered traffic signal control method for intelligent transportation systems. Their study focuses on emergency-vehicle performance while attempting to minimize negative effects on traffic in conflicting directions [3]. This research provides a basis for including emergency-vehicle priority in intelligent traffic-signal systems.

III. PROPOSED SYSTEM

The proposed system uses local processing at or near the traffic intersection. Street cameras capture traffic conditions, and a smart pole computer processes relevant visual information. The local computer can estimate traffic conditions and identify an emergency vehicle when suitable detection information is available. Based on these inputs, a traffic-signal controller selects an appropriate signal action.

The basic information flow is:

Street Cameras → Smart Pole Computer → Traffic-Signal Controller

An optional connection can forward selected information to a city cloud server for storage and wider analysis:

Street Cameras → Smart Pole Computer → City Cloud Server

The cloud component is not required for every real-time decision. Keeping the main decision process local is intended to reduce dependence on continuous communication with a remote server.

IV. EMERGENCY VEHICLE PRIORITY

The main research gap addressed by the proposed concept is rapid coordination of traffic signals for emergency vehicles. When an ambulance approaches a congested intersection, the system can use camera or other available detection information to identify the emergency vehicle and initiate a priority sequence.

Nearby signals can then be coordinated to provide a suitable green path while considering traffic in other directions. The concept is consistent with prior emergency-vehicle-centered traffic signal research [3].

However, the proposed architecture is a conceptual design and does not claim a measured improvement in emergency response time. Such performance would need to be established through simulation, field testing, or controlled experiments.

V. SYSTEM COMPONENTS

The proposed system consists of the following components:

- 1) **Street Cameras:** Capture traffic scenes and provide visual information about vehicles and road conditions.
- 2) **Smart Pole Computer:** Performs local processing of camera information and generates traffic-control inputs.
- 3) **Traffic-Signal Controller:** Receives the control decision and operates the traffic signals.
- 4) **Emergency-Vehicle Detection Module:** Identifies an emergency vehicle using an appropriate detection method.
- 5) **Optional Cloud Server:** Stores selected data and supports long-term analysis rather than being required for every immediate signal decision.

VI. EXPECTED ADVANTAGES

The proposed architecture may provide several potential advantages. Local processing can reduce the need to transmit continuous raw video to a remote server. Camera-based traffic information can support adaptive decisions rather than relying only on fixed schedules.

An emergency-vehicle priority function can allow the signal system to respond specifically to emergency situations. Centralized storage can still be used for later analysis when required.

These are expected system-level advantages of the proposed architecture; they should not be interpreted as experimentally verified performance results.

VII. LIMITATIONS AND FUTURE WORK

The proposed system requires further testing before deployment. Important issues include reliable emergency-vehicle detection, false detections, camera visibility, lighting and weather conditions, communication reliability, traffic-signal-controller compatibility, cybersecurity, and safety validation.

Future work can implement the architecture in a traffic simulation environment, compare fixed-time and adaptive control, and measure emergency-vehicle delay, queue length, waiting time, and signal efficiency.

A physical prototype can then be developed using a camera, an edge-computing board, and a traffic-light model.

VIII. CONCLUSION

This paper presented a conceptual smart traffic-light system using local AI and camera-based traffic information, with a specific focus on emergency-vehicle priority.

Existing literature supports the use of intelligent traffic-signal control, machine vision for adaptive traffic information, and emergency-vehicle-centered signal control [1]–[3].

The proposed architecture places real-time processing near the intersection while retaining an optional cloud component for storage and analysis. Future implementation and experimental evaluation are required to determine the actual performance of the proposed system.

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